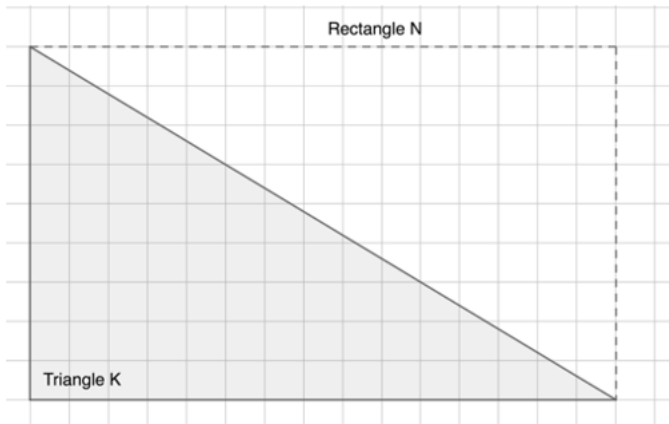


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Lesson 1 Practice

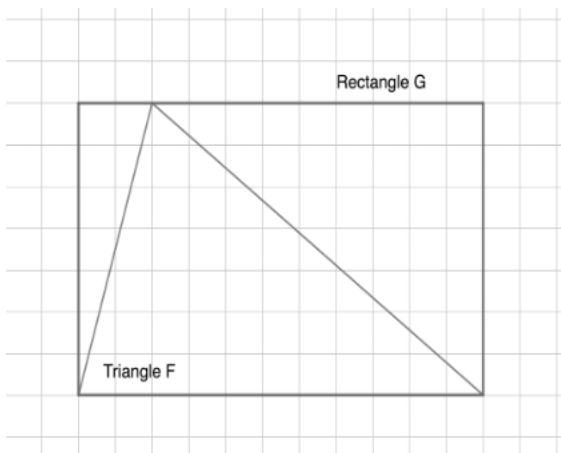
Name _____
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Triangle K is a **right triangle** within Rectangle N . Use the figure to answer questions 1-3. Each square of the grid paper is 1 cm by 1 cm.

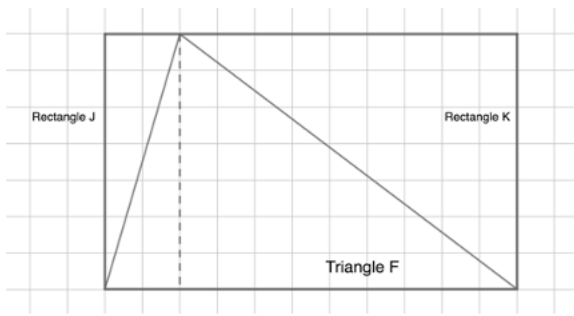


1. It takes _____ triangles to form Rectangle N .
2. The area of Rectangle N is _____ cm^2 .
3. The area of triangle K is _____ cm^2 .

The figure shows an **acute triangle**, Triangle F . Rectangle G is drawn around Triangle F .



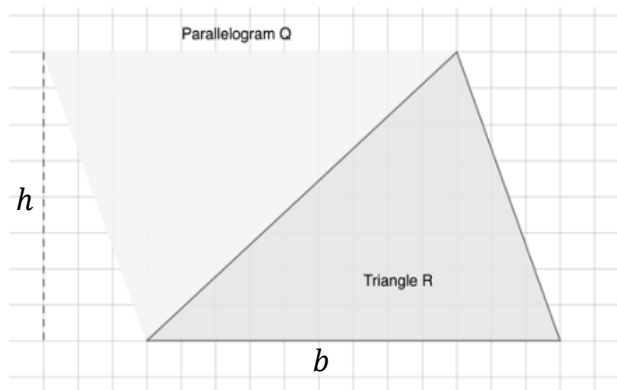
The figure can be broken into smaller rectangles, Rectangle J and Rectangle K , and 2 right triangles.



Use the figures to answer the questions 4-7. Each square of the grid paper is 1 cm by 1 cm.

4. The area of Rectangle J is _____ cm^2 .
5. The area of Rectangle K is _____ cm^2 .
6. The area of a right triangle is _____ of the area of a rectangle.
7. The total area of Triangle F is _____ cm^2 .

Triangle R is an **obtuse triangle**. Parallelogram Q is formed by duplicating Triangle R and rearranging it.



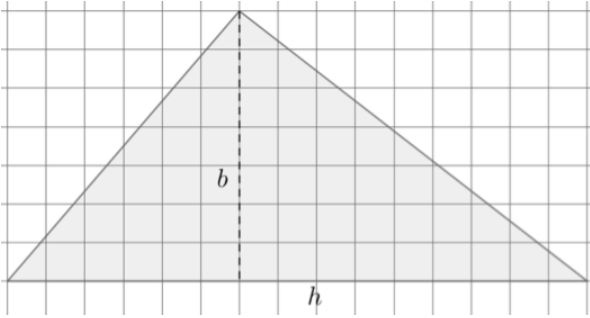
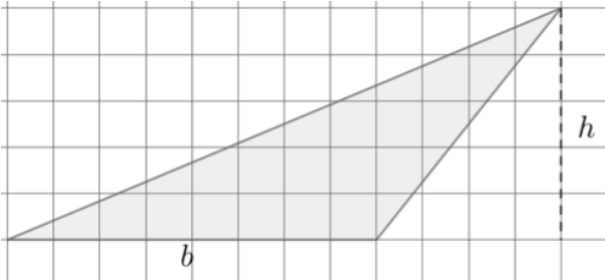
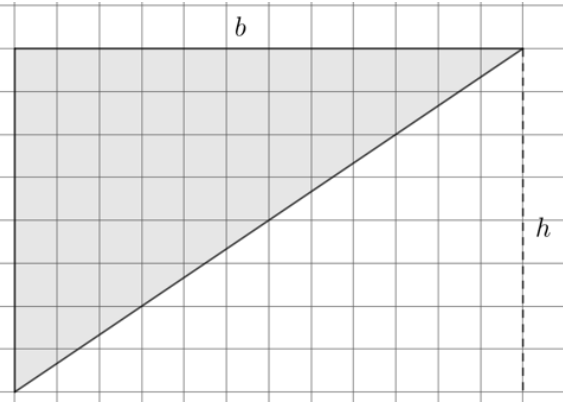
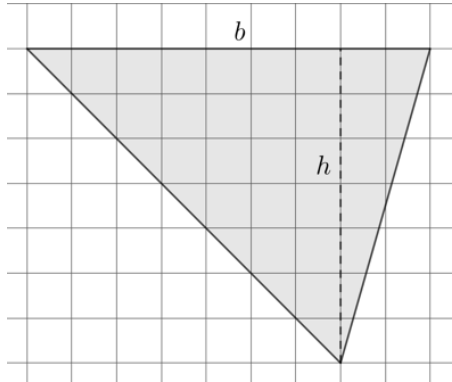
Use the figures to answer the questions 8-10. Each square of the grid paper is 1 cm by 1 cm.

8. The area of Parallelogram Q is _____ cm^2 .
9. Triangle R represents _____ of the figure of Parallelogram Q .
10. The area of Triangle R is _____ cm^2 .

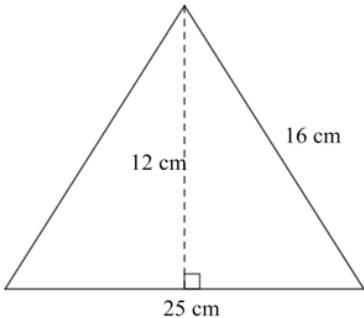
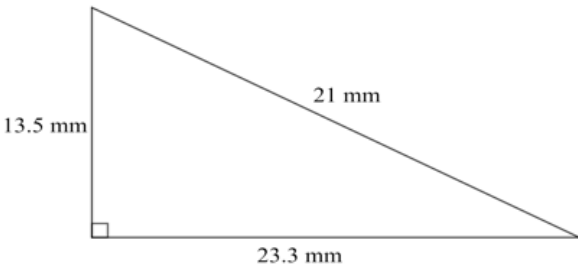
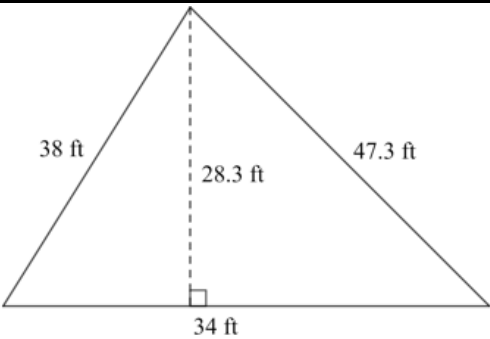
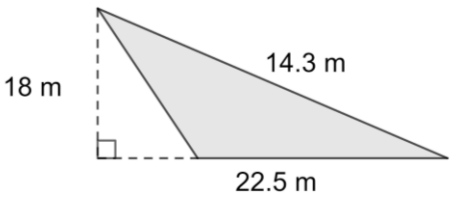
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Four triangles are shown on the grids. For each triangle, a base and its corresponding height are labeled. The side length of each square is 1 inch (in.). Find the area of each triangle. Show your work and include units.

1.		Area = _____
2.		Area = _____
3.		Area = _____
4.		Area = _____

For each triangle below, circle the base measurement and put a square around the height measurement. Then, use the base and height to find the area of each triangle. Show your work and include units.

5.		Area = _____
6.		Area = _____
7.		Area = _____
8.		Area = _____

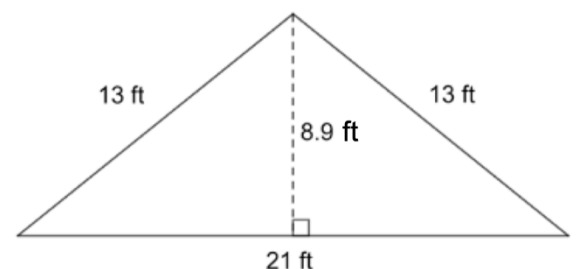
Jack found the area of the triangle shown. His work is also shown.

$$\text{Area} = 21 \times 13 \times 0.5$$

$$\text{Area} = 136.5 \text{ ft}^2$$

9. Explain why Jack's answer is incorrect.

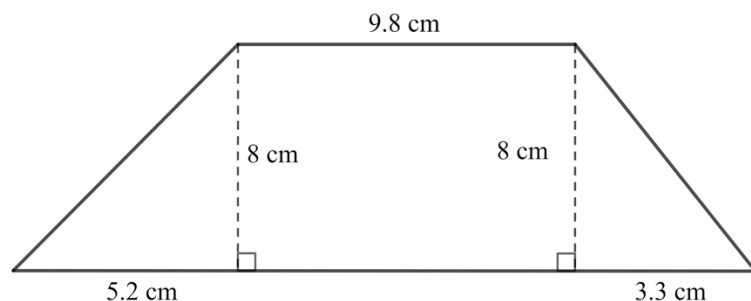
10. Find the correct area of the triangle.



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Lesson 3 Practice

Name _____
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A quadrilateral with measurements in centimeters (cm) is shown for questions 1-3.

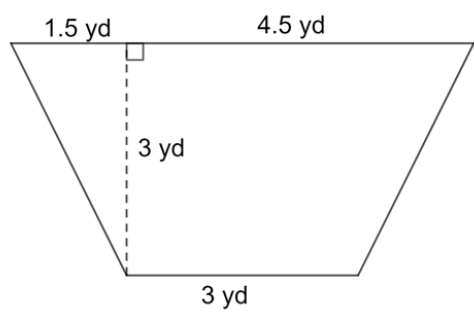
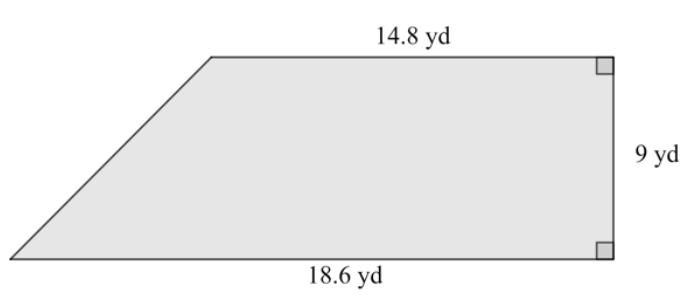


- Decompose the quadrilateral into triangles and a rectangle. Sketch the decomposed shapes here.
- Using the measurements given in the original image, label the base and height of each triangle and the rectangle.
- Find the area of the original quadrilateral by finding the sum of the areas of the decomposed triangles and rectangle. Include units.

Decompose each quadrilateral to find the area. Lengths are shown in inches (in.).

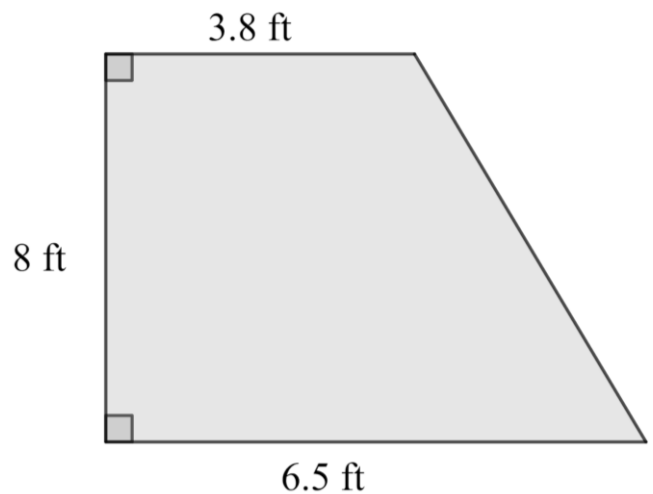
4.	5.	6.
Area = _____	Area = _____	Area = _____

Decompose each quadrilateral to find the area. Lengths are shown in yards (yd).

7. 	8. 
Area = _____	Area = _____

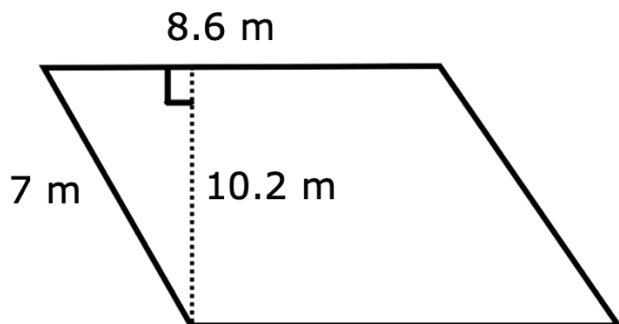
Geri made a sketch of a wall she needs to paint. The sketch of the wall is shown with measurements in feet (ft) for questions 9-10.

9. The home improvement store is offering a sale on their quarts of paint. A quart of paint covers 100 square feet of wall. Will Geri have enough paint if she buys one quart?



10. Determine the amount of paint, in square feet, Geri is short or has left over.

A parallelogram with dimensions labeled is shown below for questions 1 – 2.



1. Which dimension is not needed to find the area of the parallelogram? Explain.

2. Fill in the blanks to find the area of the parallelogram.

$$\text{Area of parallelogram} = b \cdot h$$

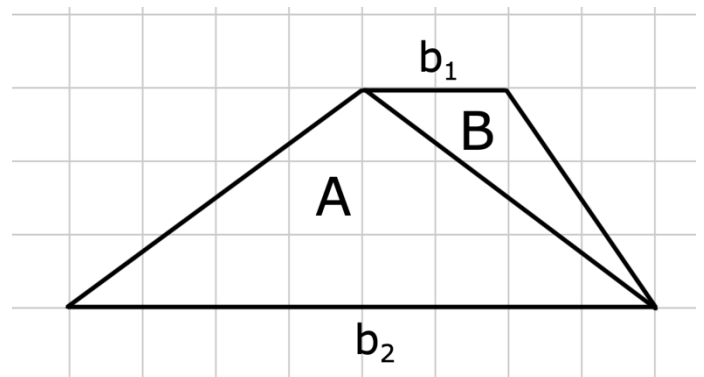
$$\text{Area} = \underline{\hspace{2cm}} \cdot \underline{\hspace{2cm}}$$

$$\text{Area} = \underline{\hspace{2cm}} \text{ m}^2$$

A trapezoid can be decomposed into two triangles with the same height. Use the diagram below for questions 3 – 6.

3. The base of triangle A, b_2 , is _____ units and its corresponding height is _____ cm. Therefore, the area of triangle A is _____.

4. The base of triangle B, b_1 , is _____ units and its corresponding height is _____ cm. Therefore, the area of triangle B is _____.



5. The area of the trapezoid is _____ units².

6. Use the formula for area of a trapezoid to verify your solution is correct.

$$\text{Area} = \frac{1}{2}h(b_1 + b_2).$$

$$\text{Area} = \frac{1}{2} \text{ ______ } (\text{ ______ } + \text{ ______ })$$

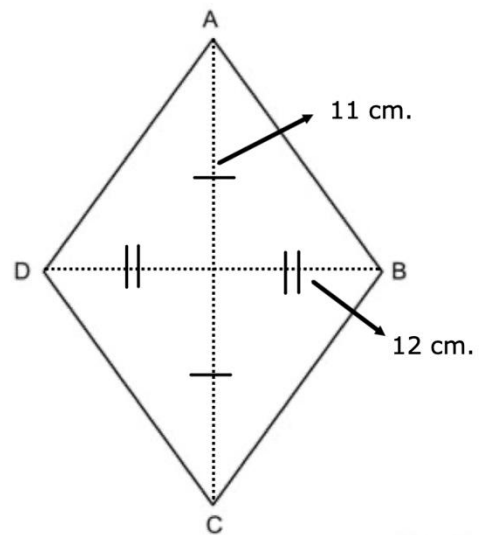
$$\text{Area} = \text{ ______ } \text{ units}^2$$

Rhombus ABCD is shown. Diagonal BD is 12 centimeters and diagonal AC is 11 centimeters. Use the diagram below for questions 7 – 10.

7. The base of triangle ABC is _____ cm. and its corresponding height is _____ units. Therefore, the area of triangle ADC is _____.

8. The base of triangle ADC is _____ cm. and its corresponding height is _____ units. Therefore, the area of triangle ADC is _____.

9. The area of the trapezoid is _____ cm^2 .



10. Use the formula for area of a rhombus to verify your solution is correct.

$$\text{Area} = \frac{1}{2}(d_1)(d_2).$$

$$\text{Area} = \frac{1}{2}(\text{ ______ })(\text{ ______ })$$

$$\text{Area} = \text{ ______ } \text{ units}^2$$

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1. The formula to find the area of a trapezoid is

- ☐ $\frac{1}{2}d_1d_2$
- ☐ bh
- ☐ $\frac{1}{2}h(b_1 + b_2)$

2. The formula to find the area of a rhombus is

- ☐ $\frac{1}{2}d_1d_2$
- ☐ bh
- ☐ $\frac{1}{2}h(b_1 + b_2)$

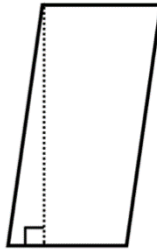
3. The formula to find the area of a parallelogram is

- ☐ $\frac{1}{2}d_1d_2$
- ☐ bh
- ☐ $\frac{1}{2}h(b_1 + b_2)$

For questions 4-6, label each figure with the given dimensions, then calculate the area using the appropriate formula.

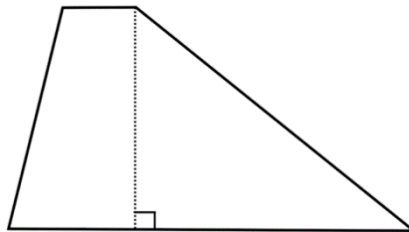
4. $b = 7.8$ in
 $h = 11.7$ in

Area =



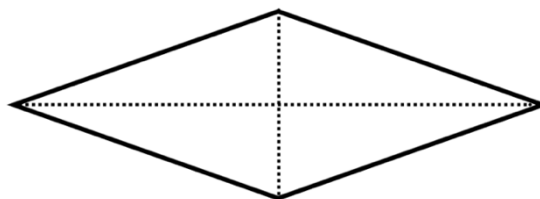
5. $b_1 = 1\frac{1}{4}$ in
 $b_2 = 6\frac{2}{5}$ in
 $h = 5\frac{2}{3}$ in

Area =



6. $d_1 = 9.5$ in
 $d_2 = 4$ in

Area =



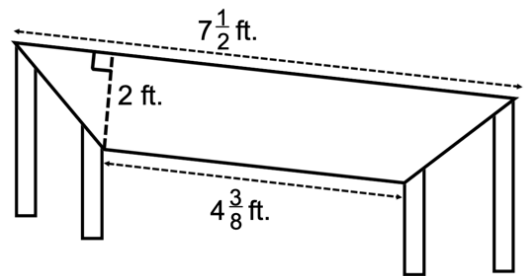
7. Jocelyn made an error in her work when solving the problem below. Identify the error and then find the base of the parallelogram.

The area of a parallelogram is 75 square centimeters and the height is 6.25 centimeters. What is the length of the base of the parallelogram?

Her solution: $(75)(6.25) = 468.75$

The length of the base is 468.75 centimeters

8. The seat of a bench in the Marlins dugout is in the shape of a trapezoid. Find the area of the bench.



9. The Adidas Flagship Building located in Herzogenaurach is shaped like a rhombus. One of the diagonals measures 255.4 meters and the other diagonal measures 203.7 meters. Find the area of the top of the building.

10. An eraser is in the shape of a parallelogram with a base of $1\frac{7}{12}$ inches and a height of $\frac{6}{14}$ inches. How many square inches is the top of the eraser?

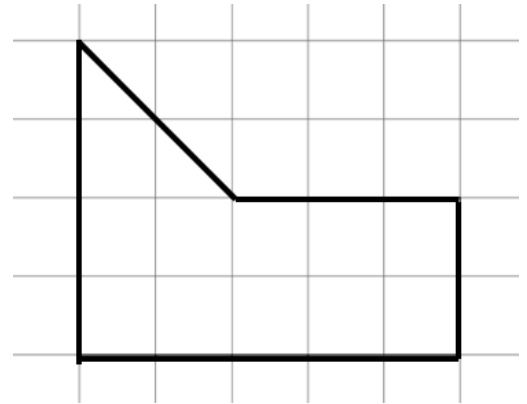


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Unit 7
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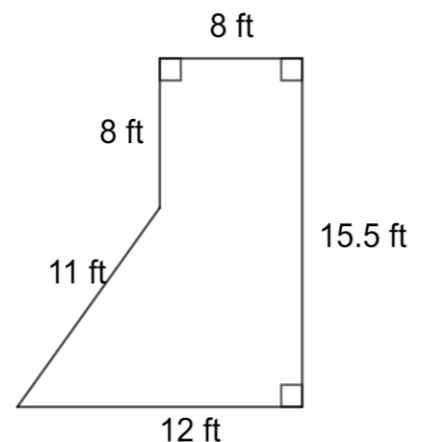
A composite figure is shown for questions 1-3. The length of each square is 1 centimeter (cm).

1. Draw lines to decompose the figure into rectangles and/or triangles.
2. Find the area of each rectangle and/or triangle. Include units.
3. Find the sum of the areas of the rectangles and/or triangles to find the area of the composite figure. Include units.



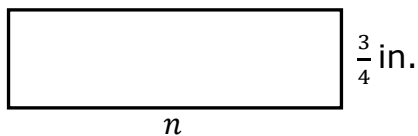
A composite figure measured in feet (ft) is shown for questions 4-5.

4. Find the area of the triangle and the rectangle. Include units.
5. Find the sum of the areas of the rectangle and triangle to find the area of the composite figure. Include units.



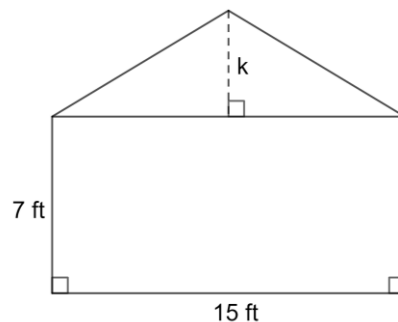
For each composite figure, determine the missing dimension.

6. Area = 4.5 in^2



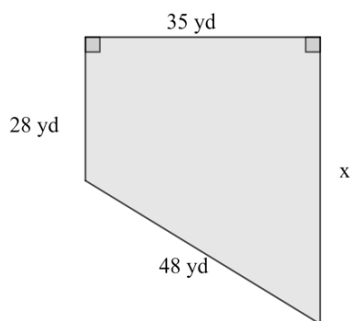
$n =$ _____

7. Area = 150 ft^2



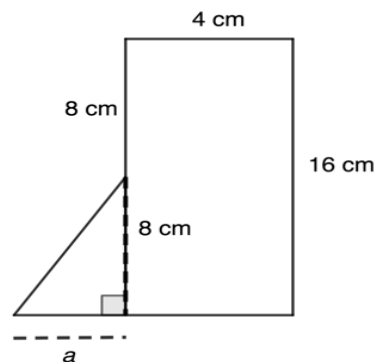
$k =$ _____

8. Area = 150 ft^2



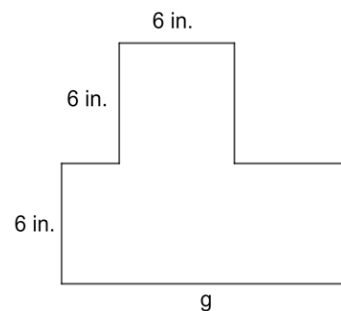
$x =$ _____

9. Area = 76 cm^2



$a =$ _____

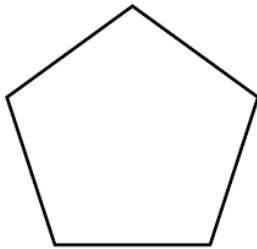
10. A composite figure with an area of $108 \text{ square inches (in}^2\text{)}$ is shown. Determine the missing dimension, g . Include units.



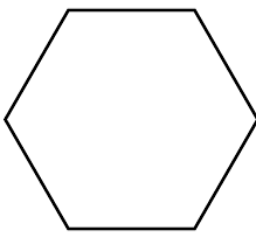
Grade 6 Accelerated
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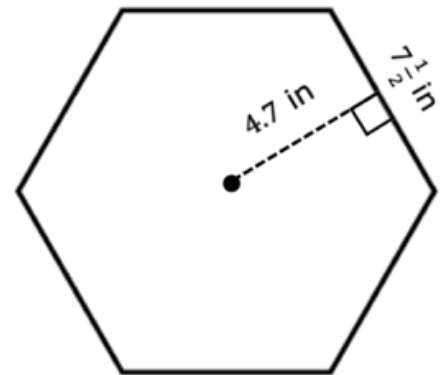
1. Decompose the regular pentagon into only triangles.



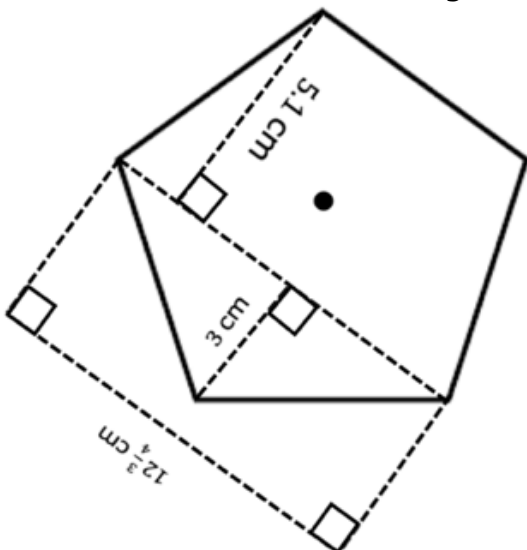
2. Decompose the regular hexagon into triangles and quadrilaterals.



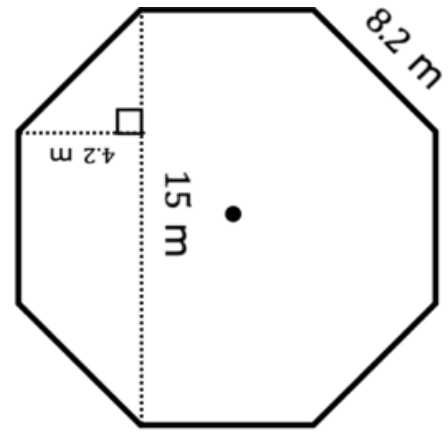
3. A hexagon is given in inches (in). Each side of the hexagon has a length of $7\frac{1}{2}$ inches and a perpendicular distance from the center to the midpoint of a side to be 4.7 inches. Find the area of the regular hexagon rounded to the nearest square inch.



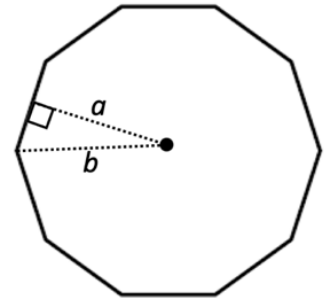
4. Find the area of the regular pentagon that has a side length of 6 cm.



5. An octagon is shown with lengths in meters (m).
What is the area of the octagon?



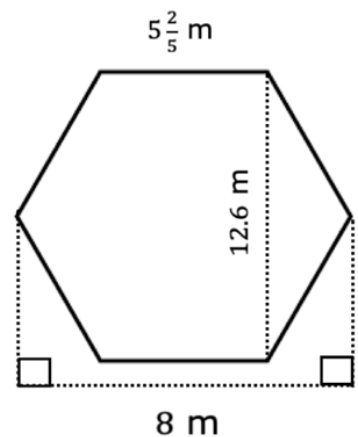
6. Label the regular decagon with the following dimensions.
Side length = $9\frac{3}{10}$ in.
 $a = 8.4$ in.
 $b = 12.54$ in.



Which of the dimensions is not needed to find the area of the decagon?

7. Find the area of the decagon from question 6 in square inches.

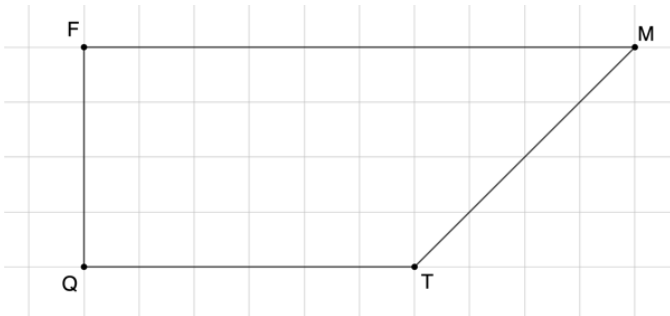
8. Mia wanted to find the area of the hexagon by decomposing it into two triangles and a quadrilateral. Help her decompose the hexagon and label any additional necessary dimensions.



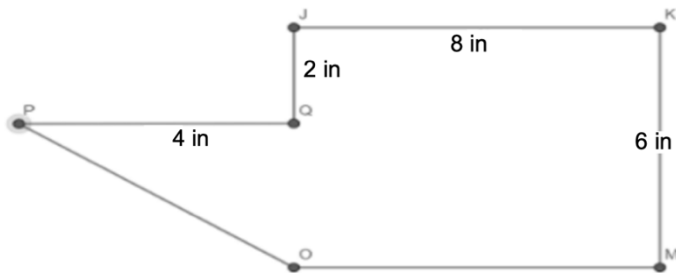
9. Find the area of the hexagon above in square meters.

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Lesson 8 Practice

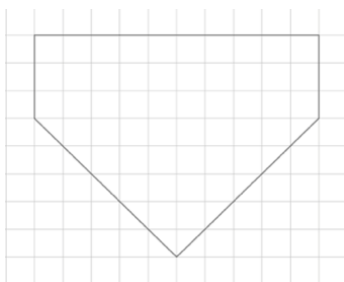
Name _____
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Period _____



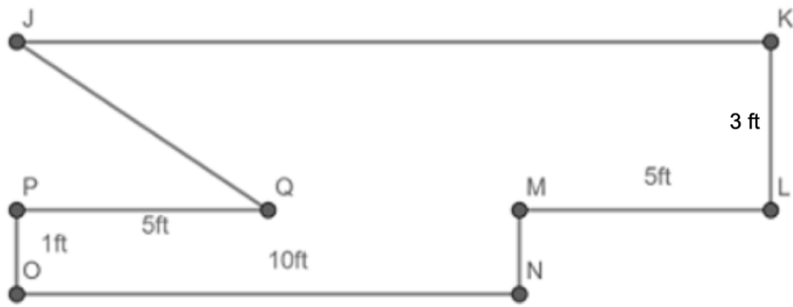
1. Decompose the composite figure above into triangles and quadrilaterals.
2. Find the area of the composite figure in question 1, if each unit represents a yard.



3. Decompose the composite figure above into triangles and quadrilaterals.
4. Find the area of the composite figure in question 3.

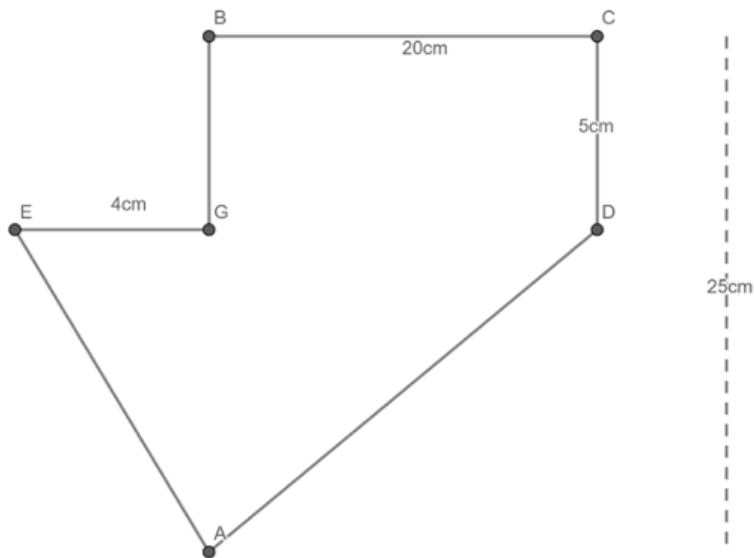


5. Decompose the image above into triangles and quadrilaterals.
6. Find the area of the composite figure in question 5, if each unit represents an inch.



7. Decompose the image above into triangles and quadrilaterals.

8. Find the area of the image in question 7.



9. Decompose the image above into triangles and quadrilaterals.

10. Find the area of the composite figure in question 9.